



Unsilenced: Using Crowdsourcing Technology to Collect and Amplify Community Voices in Climate change in Yumbe District, Uganda

April, 2025

NETHOPE

 **Ushahidi**



Executive Summary

The Climate Action Project in Yumbe District, Uganda, implemented by Ushahidi in collaboration with NetHope and YUGNET (Yumbe GBV Reduction Network), sought to enhance climate resilience and disaster preparedness in one of Uganda's most climate-vulnerable regions. Yumbe District, located in the West Nile region, experiences erratic rainfall, prolonged droughts, extreme heat, and deforestation, all of which exacerbate food insecurity, water scarcity, and environmental degradation. Additionally, the presence of Bidibidi Refugee Settlement, one of the world's largest refugee camps, has intensified competition for resources between host and refugee communities.

The project leveraged Ushahidi's crowdsourcing technology to collect and analyze community-driven climate risk data. A participatory approach was employed, reaching 1,752 community members across 14 sub-counties through SMS, WhatsApp, and in-person interviews through the Ushahidi mobile app. These insights were further validated through stakeholder engagement and community consultations to ensure that the findings reflected the lived realities of both host and refugee populations.

The results revealed that unpredictable rainfall patterns, prolonged dry spells, and rising temperatures are key climate stressors disrupting agricultural productivity and water availability. Communities reported that low crop yields, soil erosion, and deforestation have worsened economic hardships, forcing some households to rely on alternative livelihoods, such as charcoal burning, casual labour, and the sale of fast-growing tree species. Social vulnerabilities, including gender-based violence and food insecurity, were also linked to the worsening climate conditions.

Through the validation sessions, community-driven recommendations were developed and categorized into response, adaptation, and mitigation measures. These included strengthening early warning systems, investing in water harvesting infrastructure, distributing drought-resistant crop varieties, promoting alternative energy sources, and implementing reforestation programs. The report also highlights the need for inclusive governance, enhanced disaster preparedness, and stronger enforcement of environmental policies to ensure long-term resilience.

This report provides a comprehensive overview of climate risks, socio-economic impacts, and community-led adaptation strategies in Yumbe District. It serves as a valuable resource for policymakers, humanitarian organizations, and local communities, offering actionable recommendations to enhance climate resilience and improve disaster response mechanisms in the region.

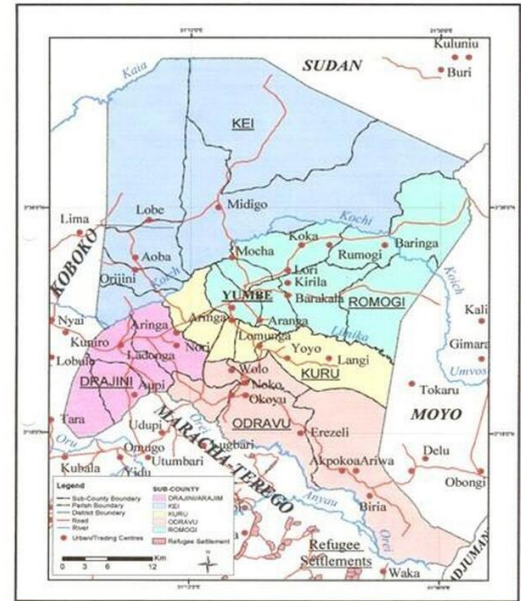
Key Findings

- *Community Perceptions of Climate Risks:* Unpredictable weather patterns and deforestation are worsening economic hardships, food insecurity, and social vulnerabilities.
- *Climate Change Impacts on Livelihoods:* Reduced crop yields, extreme weather events, and rising temperatures are severely affecting agriculture, income sources, and food security.
- *Community Adaptation Strategies and Challenges:* Some adaptation strategies, such as irrigation and food storage, are in place, but limited resources hinder widespread implementation.
- *Disaster Preparedness and Governance:* Weak disaster response systems and limited enforcement of environmental regulations are exacerbating climate-related vulnerabilities.
- *Public Confidence in Climate Response Organizations:* NGOs are more trusted than government agencies in addressing climate challenges, but stronger collaboration is needed.
- *Community Priorities for Climate Resilience:* The community prioritizes reforestation, water conservation, early warning systems, and inclusive participation in climate governance.
- *Recommended Climate Action Measures:* Strengthen early warning systems, promote climate-smart agriculture, improve water management, and invest in clean energy solutions.

Introduction

Location and Administrative Units

Yumbe District is located in the West Nile region of Uganda. The district borders the Republic of South Sudan to the North, Moyo District to the East, Obongi District to the South East, Koboko District to the North West, Maracha and Terego to the West, and Madi-Okolo District to the South West. Yumbe District covers a total surface area of 2,411sq. km (which is 1.2% of the total national area), 1,929 sq. km (80.01%) of the area is arable land under agriculture, 411.78 sq. km (17.08%) Forestry and woodlands while water bodies and wetlands account for 70.22 sq. km (2.912%). It consists of 14 sub-counties and is home to one of the largest refugee settlements in the world - Bidibidi Refugee Settlement. Administratively, the district is governed by a local council system, with decentralized leadership structures managing development programs across sub-counties and parishes.



Topographic and Drainage Features

Yumbe District features a predominantly flat terrain interspersed with rolling hills in the eastern part, and in the north, there are two hills, namely Midigo and Kei, which play a vital role in water retention during the rainy season. The district is drained by seasonal rivers and streams, most of which flow into the River Nile. However, many of these watercourses dry up during prolonged droughts, intensifying water scarcity challenges. Soil types in Yumbe vary, with loamy soils in the lowlands that support agriculture and sandy soils in the uplands, which are prone to erosion during heavy rains.

Climate and Ecological Conditions

The district experiences a tropical savanna climate characterized by distinct wet and dry seasons. Rainfall averages between 800 mm and 1,200 mm annually, but it is increasingly erratic due to climate change. The long rains typically occur between April and May, while the short rains fall from September to November. Temperatures range from 20°C to 35°C, with the hottest months being December to February. Vegetation in Yumbe includes savanna grasslands and scattered woodlands, but deforestation and overgrazing have significantly reduced tree cover, exacerbating soil erosion and desertification.

Population Size and Composition

According to the 2024 National Population and Housing Census conducted by the Uganda Bureau of Statistics, Yumbe District's population has increased to 945,100 people, making it the most populous district in the West Nile subregion. The Bidibidi Refugee Settlement, located within Yumbe District, hosts approximately 198,549 refugees as of April 30, 2024, primarily from South Sudan. This refugee population constitutes about 21% of the district's total population. Demographically, Yumbe has a high proportion of youth, with 68% of the population under 25 years old, leading to a significant dependency ratio and limited workforce capacity.

Socio-Economic Context

Yumbe District's economy is heavily reliant on subsistence agriculture, with approximately 80% of its total area dedicated to this practice. The main food crops grown are maize, sesame, cassava, sweet potatoes, millet, sunflower, sorghum, beans, honey, shea butter, tobacco, cotton and mangoes. However, limited access to markets, poor infrastructure, and frequent climate shocks undermine economic productivity. Among refugees, subsistence farming in allocated plots and casual labour are common income sources, though land access and resource competition with host communities often create tensions. Dependency on humanitarian aid is widespread, particularly among refugees, as opportunities for formal employment remain scarce. Tobacco is the major cash crop and main source of livelihood for many in the district. It is grown mainly in the fertile highland areas. The people also engage in general retail and wholesale, metal and wood fabrication, arts and crafts production, fish farming and livestock farming.

Yumbe District Disaster Profile

Climate change in Uganda, including in West Nile, materialises in an increase in average temperature and heatwaves, more unpredictable rainfall, and increased crop and land exposure to drought. Over the past 20 years, especially the northern regions of Uganda have experienced more frequent and longer-lasting drought conditions. The livelihoods and food security of people in West Nile are highly dependent on rain-fed subsistence farming, which is vulnerable to shocks and adverse weather events related to climate change and climate variability. Droughts in West Nile have already resulted in scarcities in local food markets and inflated the prices of food staples. Even people with access to normally fertile lands have faced severe climate-driven variability in recent years, contributing to increasing livelihood insecurity.

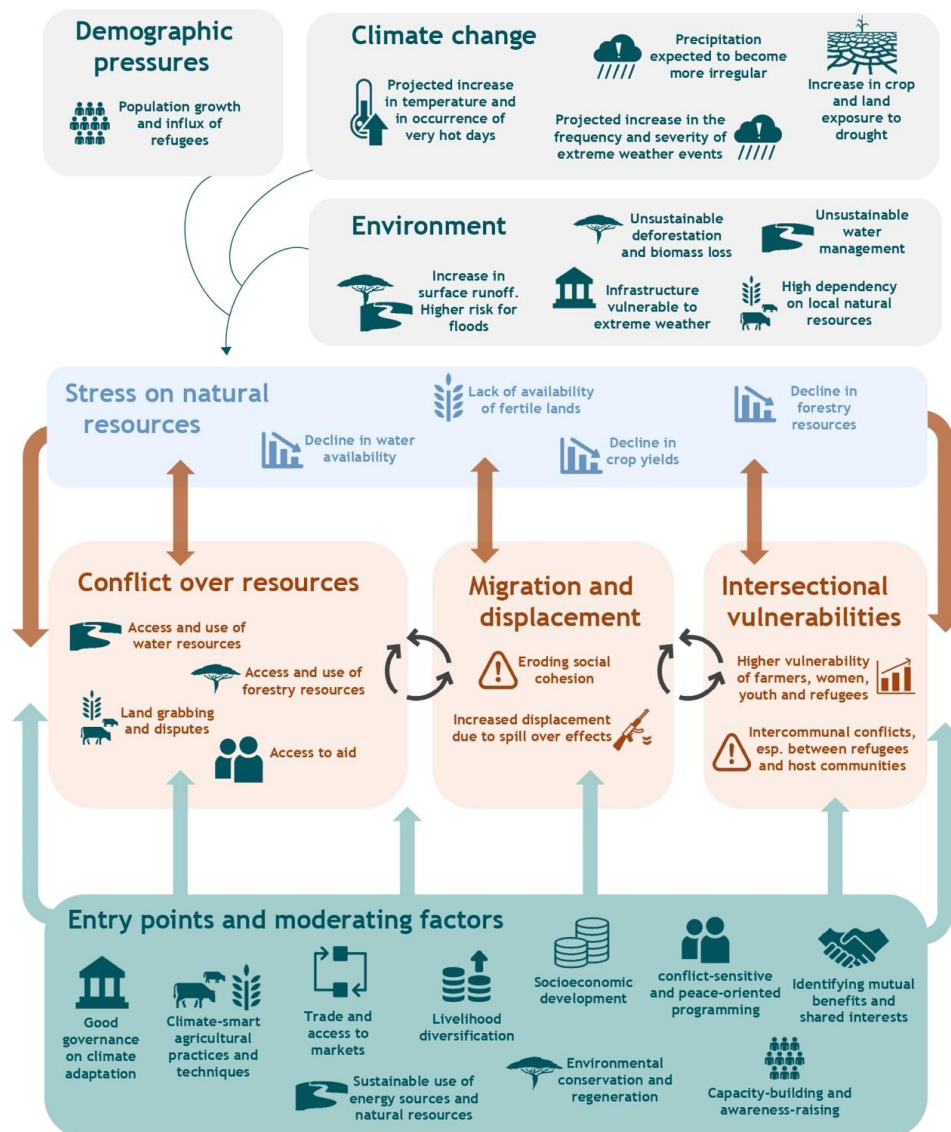
In addition to these climate change impacts, the influx of refugees has put the local environment further under strain. In particular, forestry, agricultural land, and water resources are scarcer and more difficult to access. Fertile land is an important but traditionally sensitive resource in West Nile, essential for cultural identity and for sustaining livelihoods and the local food basket. Land ownership in West Nile also has a complicated history, marred by conflict and political interference. The Ugandan authorities have negotiated with local community leaders to acquire land for use by refugees. However, in some cases, the allocation of portions of land to refugees caused resentment amongst the local population and intercommunal tensions.

Forestry resources in West Nile are also a source of tension and potential conflict. Trees are being cut down at an unsustainable rate due to the spike in demand for energy sources driven by the rapid population growth, but also because the vast majority of refugee and host population households still rely on wood as primary cooking fuel. This problem is particularly critical in rural areas and around refugee settlements, where forestry management is weak and unsustainable. A scarcity of forestry products increases the risk of tensions and conflict around the competition to control and access these resources. It also heightens the risk of sexual- and gender-based violence, as women and girls are primarily responsible for collecting firewood, and a lack thereof means they have to spend more time and cover greater distances in this process.

Further, per capita water availability in West Nile is expected to continue to decline due to population growth, the effects of climate change and poor water infrastructure. While conflicts over access and availability of water in West Nile are still rare, communal competition and small-scale conflicts over water points and natural springs are already occurring. As most of the population in West Nile is dependent on rain-fed subsistence farming, a further decrease in water availability will directly impact people's livelihoods and food security, risking water becoming more salient in conflict dynamics.

Finally, West Nile also experiences risks related to climate - and conflict-induced migration and displacement. Firstly, further deterioration of the security situation in South Sudan could result in a renewed influx of refugees in West Nile, leading to increased pressures on livelihoods and food security, potentially driving communal conflict. On the other hand, more stability in South Sudan may result in out-migration as refugees decide to return home, which could relieve some of the current pressures on land, forestry, and water resources. In the mid to longer term, changes in internal migration patterns and trends in Uganda could lead to increased demographic pressures, including in West Nile. For instance, when the arid north-eastern regions of Uganda become less suitable for livestock rearing due to climate change, the more fertile north-western regions could serve as better alternatives for herders in search of grazing lands. Poorly managed seasonal migration could quickly lead to transhumance-related conflicts, as seen in other regions in sub-Saharan Africa. In contrast, accelerated deforestation and further decreases in water availability and crop production in West Nile could also lead to major out-migration hotspots in the future. Finally, such climate-induced human mobility could lead to a population spatial shift from rural to urban areas, causing demographic pressures in regional towns.

Interplay of Climate Security Risks in West Nile



Project Objectives and Methodology

The primary goals of the project were to:

- **Partner with NetHope to update current knowledge on what community-level climate risks significantly impact livelihoods in the Yumbe District.** These communities are currently exposed to frequent and intense events like droughts that have led to significant agricultural loss and, subsequently, loss of livelihoods and reduced productivity.
- **Co-create low-tech and participatory data collection tools to gather reports from at least 1000 residents of Yumbe District in compliance with the data protection and privacy obligations of Uganda.** Working with the communities gave them agency and a sense of ownership over the information generated. This increased the relevance and accuracy of the data because community members have firsthand knowledge of their local context, including climate risks, livelihood practices, and vulnerabilities.
- **To assess the community's perception of risks associated with past and current climate variations.** The Yumbe District community provided insights into the strategies that have worked or failed in the past and proposed innovative ideas to address the challenges they face. This participatory approach ensures that interventions and policies will be tailored to the community's socio-economic needs, increasing the likelihood of their success.

These objectives were geared towards achieving the outputs below:

- **Climate risk maps created for Yumbe District.** These climate risk maps revealed new opportunities for, e.g., economic diversification and innovation. By understanding the risks associated with traditional livelihoods, communities can explore the relevant alternative income sources and business models that are less vulnerable to climate impacts. This may include investing in renewable energy, eco-tourism, climate-resilient agriculture, or other sectors that align with the community's strengths and resources.
- **Visualized community reports and timelines of climate-induced shocks, coping mechanisms, and local resource maps.** These reports have provided crucial information for local financial institutions and insurance companies to assess risks and offer appropriate financial products. A clear knowledge of climate risks and socio-economic information will enable stakeholders delivering financial services, such as loans and insurance, to specifically design their offerings to address climate-related vulnerabilities and enhance economic resilience and recovery in Yumbe District.
- **Validation and dissemination of local-level climate adaptability and resilience strategies.** Ushahidi's validation approach used focus group discussions for vulnerable groups and town hall sessions for key stakeholders and local communities. These allowed for direct feedback and two-way engagement. Our products, after validation, included an interactive Ushahidi website deployment for anyone to explore the climate risk maps for Yumbe District, including socio-economic opportunities for adaptability and resilience in response to the effects of climate change; a case study publication that documented our methodology and data sources, and showcased climate risk maps, with context-specific analyses of the opportunities for economic diversification and innovation—for researchers and stakeholders wanting in-depth information; and a concise policy brief including infographics and key insights for local leaders and policymakers. To bring awareness to these products and encourage responsive actions, Ushahidi engaged with community members, relevant partners, and stakeholders through dissemination workshops, dialogues and social media to reach a wider audience.

Key Findings

Community Perceptions of Climate Risks and Adaptation Strategies and Collaborative Development of Low-Tech, Participatory Data Collection Tools for Engaging Yumbe District Residents

In May, a discovery session was conducted at Escape Village Hotel in Yumbe Town, and a meeting with district governance at the Yumbe district office to engage a diverse range of community members, including women, youth, persons with disabilities, and local leaders. This session aimed to ensure that the perspectives of both host communities and refugees were incorporated into the design of the data collection process to be implemented through the Ushahidi Platform. The discovery session served as an introduction to the project, its objectives, and its commitment to addressing climate-related challenges in Yumbe District. Participants were encouraged to share their lived experiences, which provided critical insights into key areas such as:

General Climate Trends

Participants in Yumbe District noted significant changes in weather patterns over the past five years. Key trends included prolonged and merged rainfall seasons (e.g., from January to July, compared to the traditional February to April), unpredictable rainfall, and extended periods of intense heat. These changes have disrupted traditional farming cycles, reduced crop yields, and led to widespread economic strain. Deforestation, particularly near wetlands, was identified as contributing to reduced vegetation cover, exacerbating the district's vulnerability to climate change impacts.

Natural Disasters and Environmental Hazards

Frequent droughts and flooding emerged as the primary natural disasters affecting Yumbe District. Droughts have resulted in livestock deaths, reduced agricultural productivity, and economic decline, while flooding, particularly near riverbanks, has displaced families and destroyed critical infrastructure. Additionally, Participants highlighted several human activities exacerbating environmental degradation in Yumbe District. Large-scale logging, particularly of African trees, which take up to 100 years to mature, has significantly reduced tree cover. These trees, valued for their timber, are being cut down at alarming rates for economic gain. Further, increased mining practices, such as gold mining at Oldravu West, have led to severe land degradation and ecosystem disruption, further undermining the district's resilience to climate impacts.

Impact on Community Livelihoods

Economic activities in Yumbe District are diverse but heavily influenced by climate variability and resource availability. The primary activities include small-scale farming, livestock keeping, and mining and quarrying. Charcoal burning is also widespread, providing income for many but contributing to significant environmental degradation. Other economic activities include small businesses such as craftsmanship and shopkeeping. Additionally, vices such as betting and mugging were identified as supplementary economic activities for some community members, reflecting the strain on traditional livelihood sources caused by climate change.

Disaster Preparedness and Governance

Participants emphasized the lack of adequate disaster response plans and early warning systems. Although some NGOs and local government initiatives, such as DRDIP (Development Response to Displacement Impacts Project), are promoting tree planting and awareness campaigns, gaps in enforcement of by-laws against charcoal burning and logging remain significant. Coordination between government and NGOs was deemed crucial to improving disaster preparedness and governance.

Confidence in Response Organizations

Community members expressed varying levels of confidence in organizations addressing climate-related challenges. NGOs such as the Red Cross, WFP, Save the Children, World Vision, United Nations, and the International Rescue Committee (IRC) were the most known and trusted/preferred. Government agencies, including the Office of the Prime Minister and local government disaster management committees, were acknowledged for their involvement but received less confidence due to perceived inefficiencies. Overall, participants indicated a stronger preference for NGO-led interventions, citing their efficiency and hands-on approach in addressing the district's climate and socio-economic challenges.

Public Participation

Community members expressed a willingness to participate in climate resilience initiatives but noted challenges in inclusion and representation. Women, youth, and persons with disabilities reported limited involvement in decision-making processes. Existing structures, such as the District Disaster and Response Management Committees, require greater inclusivity to ensure all voices are heard and represented.

Community Priorities

Key priorities identified by the community include strategies to combat deforestation, enhance water resource management (e.g., boreholes and rainwater harvesting), and introduce early warning systems. Participants also emphasized the need to bridge the knowledge gap on the importance of community participation in climate action. Efforts to combat corruption, which undermines local by-laws and environmental protection, were highlighted as critical.

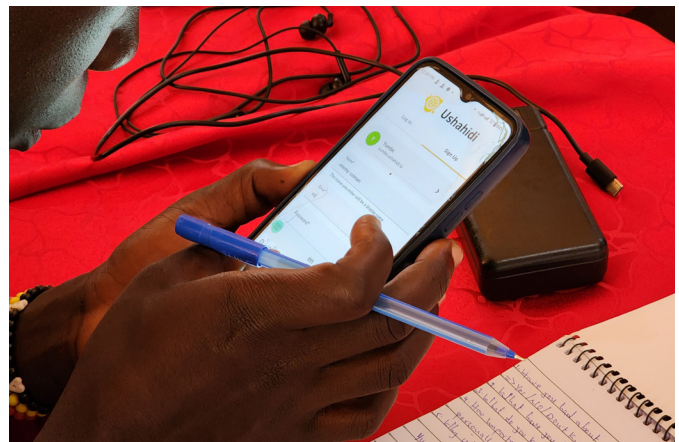
Communication Channels and Timing

Participants identified local radio stations (e.g., Ribat Radio FM and Arua One FM), community barazas, and social media platforms like WhatsApp as effective communication channels. Mobile network coverage, however, remains a challenge in some areas, with MTN and Airtel being the predominant providers. Young people, who frequently use mobile phones, were identified as a key demographic for engaging in climate action initiatives.

These insights were instrumental in shaping the design of low-tech, participatory data collection tools tailored to Yumbe District's unique needs and context. By engaging the community directly, the project ensured that the survey questions reflected their lived realities, addressing both the causes and effects of climate-related challenges. This approach underscores the importance of collaboration and inclusivity in creating solutions that are not only data-driven but also community-informed.

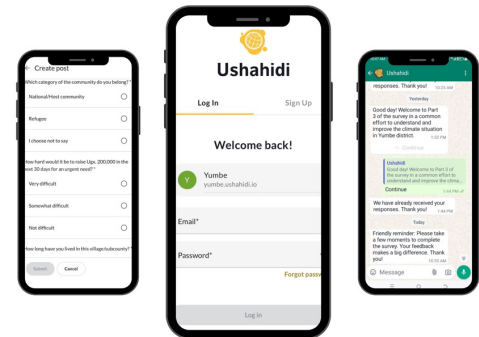
Thus, the project employed a multi-faceted data collection approach

- Community Surveys: Data was gathered from 1,752 residents across 14 sub-counties using SMS, WhatsApp, and face-to-face interviews. These methods ensured accessibility for both host communities and refugees.
- Stakeholder Validation Meetings: To validate the findings, a follow-up session was held with local authorities, community leaders, NGOs (a Livelihood Officer from the Adventist Development and Relief Agency (ADRA) and a World Vision officer involved in the greening program in Yumbe) and representation from the community. Real-time feedback was incorporated to refine the accuracy and relevance of the data collected.
- Engagement of Data Champions: Local data champions played a pivotal role in facilitating community interactions and collecting data. Their involvement helped build trust within the communities and encouraged widespread participation in the project.



Community Reports on Climate Risks

Yumbe District faces growing climate-related challenges that are exacerbating socio-economic vulnerabilities among both host communities and refugees. This section highlights findings gathered through participatory data collection across the sub-counties using the Ushahidi Platform via SMS and WhatsApp from July to September, as well as community data champions. Thus, critical challenges, adaptation needs, and community perceptions of climate change impacts.



Climate Change Impacts

Erratic Weather Patterns and Crop Failures

- Community members frequently described weather unpredictability as a major challenge. They noted that “too much sunshine in the first season and too much rain in the second season” disrupt traditional farming cycles/seasons.
- Others reported a complete drop in rains or shifts in rainfall patterns that have changed planting periods, reducing agricultural yields and causing widespread food insecurity.
- For many households, meals are eaten only once a day due to low crop production, reflecting the severity of the crisis.

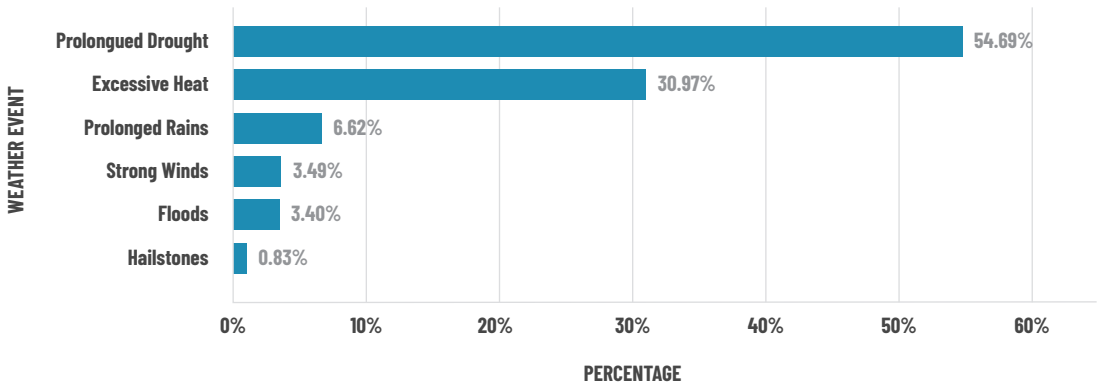
Flooding and Infrastructure Damage

- Frequent heavy rains and strong winds have led to significant destruction. One community member shared that their “house collapsed due to heavy rain and winds,” while others reported difficulty accessing basic needs due to the high prices of essentials in the market.
- Floods also destroy crops and isolate communities, further straining livelihoods and access to services.

Extreme Heat and Drought

- Rising temperatures were a recurring theme in community feedback, with extreme heat causing physical challenges. Others mentioned cracking of feet, red-eye diseases, and hunger as direct consequences of prolonged heat exposure. Some households have resorted to sleeping with their windows open during the dry season, leaving them vulnerable to theft.

Which is the main weather event that affects your household most?

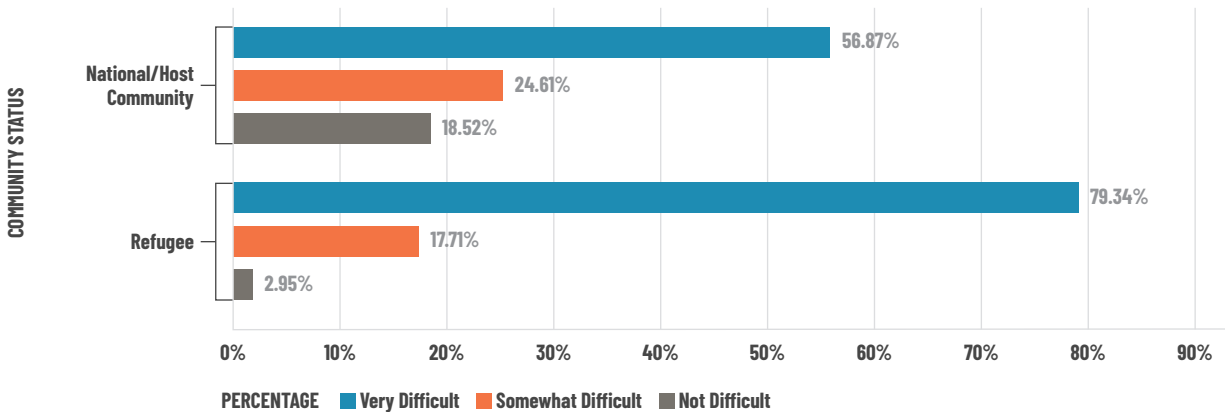


Socio-Economic Impacts

Livelihood Disruptions

- Climate change has severely impacted agricultural livelihoods, which form the backbone of the district’s economy. Community members shared that low crop yields have left them with low savings due to a lack of income. Financial strain is compounded by increased operational costs and reduced revenue for small businesses as community members reported that extreme heat damages goods in kiosks, leading to financial losses.

How hard would it be to raise Ugx. 200,000 in the next 30 days for an urgent need? (by community status)



Gender-Based Violence and Social Strain

- Community members highlighted that food scarcity has intensified household tensions, leading to increased gender-based violence (GBV). These social challenges exacerbate the vulnerability of women and children, who already face a disproportionate burden in coping with climate-related stress.

Health and Nutrition Challenges

- Community members highlighted issues such as hunger, physical strain, and heat-related challenges. Reports mentioned “cracking of feet,” “red-eye diseases,” and exhaustion caused by extreme heat and long periods of drought. These conditions contribute to additional socio-economic stress, as families face increasing medical needs and reduced ability to work.

The findings reveal a complex interplay between climate change impacts and socio-economic challenges. Recurrent droughts, flooding, and health crises are eroding resilience across Yumbe District, necessitating urgent, inclusive, and sustainable interventions. By identifying these critical areas of concern, the project lays the groundwork for informed, community-driven responses that address both immediate and long-term needs.

Validation and Dissemination of Local-Level Climate Adaptability and Resilience Strategies

The validation and dissemination phase, which took place in November, focused on ensuring that the data collected accurately reflected the lived realities of the community members and aligned with local priorities. This phase was integral to building trust, fostering ownership of the findings, and creating a foundation for actionable, community-driven solutions.

The validation session was held at Escape Village Hotel in Yumbe Town Council and engaged a diverse group of community members, including women, youth, local leaders, etc. Through structured discussions and interactive feedback mechanisms, the session provided a platform for community members to review and confirm the accuracy of the data collected during the data collection phase of the project.

Local government representatives and community leaders were also involved in the validation process. They reviewed the findings, emphasizing the importance of addressing pressing issues such as food insecurity, environmental degradation, and social challenges like increased gender-based violence linked to resource scarcity. Stakeholders acknowledged the value of aligning these findings with ongoing climate adaptation strategies and district-level plans to enhance community resilience.

The validation and dissemination process not only strengthened the accuracy and relevance of the project's findings but also created a sense of ownership among the community and stakeholders. By involving local actors at every stage, the project laid a strong foundation for the development of tailored, actionable resilience strategies that reflect the unique socio-economic and cultural dynamics of Yumbe District. This collaborative approach ensures that future interventions are rooted in the lived realities of the people they aim to support, increasing their effectiveness and sustainability.



Community Feedback and Recommendations

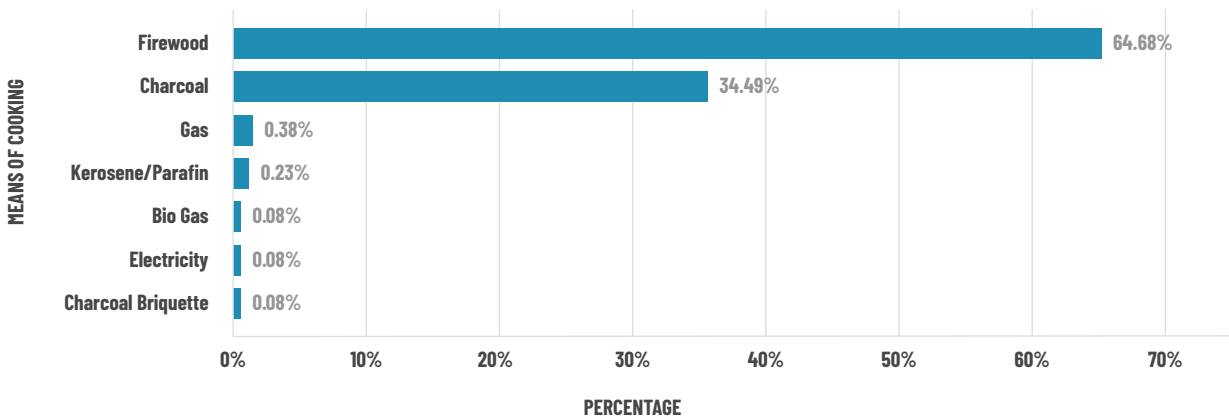
The project’s validation session provided an invaluable platform for community members to propose actionable solutions to mitigate the impacts of climate change in Yumbe District. Their recommendations have been categorized into response, adaptation, and mitigation measures to address immediate needs, build resilience, and reduce long-term vulnerabilities.

1. Community Feedback on Environmental Challenges

Means of Cooking

- **Primary Fuels Used:** Households in Yumbe District predominantly rely on firewood and charcoal for cooking. These energy sources are widely used across both host communities and refugee settlements due to their accessibility and affordability.
- **Potential Environmental Impacts:** The extensive reliance on firewood and charcoal may contribute to deforestation in areas where these resources are extracted. Community members expressed concern about the availability of firewood, which they reported is becoming harder to access, possibly due to overharvesting.
- **Possible Health Challenges:** Traditional cooking practices using open fires and inefficient stoves could lead to indoor air pollution, which may pose risks to respiratory health.
- **Economic Considerations:** The increasing demand for and reliance on charcoal may drive up its cost, adding financial strain to vulnerable households. Community members emphasized their limited ability to afford alternative energy sources, reflecting economic pressures tied to their cooking needs.

Means of Cooking



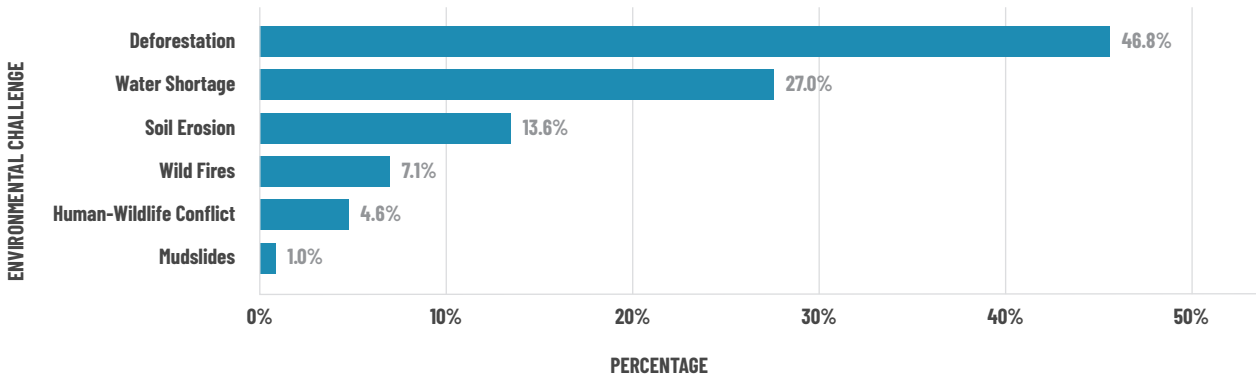
Temperature and Rainfall Changes

- **Increased Temperatures:** A majority of respondents confirmed that temperatures in Yumbe District have increased over time. As a result of these rising temperatures and declining rainfall, prolonged drought has become the main weather event affecting households. Respondents also highlighted that the intensity of these weather events has grown over the years and now occurs annually. These recurring and increasingly severe events have significantly reduced water availability, decreased agricultural productivity, and heightened the vulnerability of households that depend on farming for their livelihoods.
- **Unpredictable Rainfall Patterns:** Respondents described erratic rainfall patterns, with prolonged dry spells and sudden heavy rains. These shifts have delayed planting seasons, disrupted traditional farming schedules, led to reduced crop yields and food shortages, and accelerated soil erosion due to intense rainfall, further decreasing agricultural land productivity. Despite the heavy rains, respondents observed that it has been raining less and less over the years, pointing to a long-term decline in overall rainfall. This reduction exacerbates water scarcity, affects the replenishment of water sources, and places additional strain on agricultural systems already grappling with changing climate conditions.

Other Environmental Challenges

- **Deforestation:** Community members highlighted deforestation as the primary other environmental issue they face. The widespread cutting down of trees, primarily for charcoal and firewood, has led to significant environmental degradation in Yumbe District.
- **Water Scarcity:** Community members noted challenges related to water scarcity, affecting both household needs and agricultural activities, and;
- **Soil Erosion:** The loss of tree cover has exacerbated soil erosion, affecting agricultural productivity and increasing the risk of land degradation.

Which other environmental challenge do you face?



Effects of Environmental Challenges on Sources of Income

- **Financial Strain:** Many individuals reported significant financial difficulties due to reduced earnings from agriculture, as unpredictable weather patterns and prolonged droughts have led to poor crop yields.
- **Crop Damage and Loss:** Extreme weather conditions, such as excessive heat and irregular rainfall, have severely impacted crops, causing losses that affect both household food supplies and income from selling produce.
- **Alternative Income Sources:** Some community members have started exploring alternative income-generating activities, such as selling fast-growing tree species, to cope with reduced agricultural productivity.
- **Business Challenges:** High temperatures have also caused damage to goods in small businesses, resulting in unsold products and further financial setbacks for local entrepreneurs.
- **Low Savings:** Individuals face challenges in saving money due to reduced income, making it difficult to participate in savings programs.

1. Community Adaptation Measures

- **Sustainable Agricultural Practices:** In response to changing weather patterns and reduced rainfall, some community members have started planting crops in the second season, adopting irrigation systems, and optimizing grazing practices such as rotational grazing. However, many still lack access to resources and knowledge needed to fully implement these strategies.
- **Food Security Strategies:** While some households have planted, harvested, and stored non-perishable food crops like beans and maize, others continue to struggle with food shortages. To stretch limited supplies, some families have resorted to reducing the amount of food consumed during dry seasons.
- **Environmental Conservation and Energy-Saving Practices:** A few community members have begun planting trees and implementing energy-saving measures such as reducing electricity use and adopting solar energy. However, broader adoption remains limited, and deforestation continues to be a challenge.
- **Community Engagement and Awareness:** While some individuals have joined climate-related community initiatives, participation remains low. Many community members still lack access to timely weather forecasts and climate information, limiting their ability to proactively adjust their activities.
- **Soil Conservation and Erosion Control:** To prevent soil erosion, a portion of the community has adopted techniques such as mulching, terracing, and building barriers. However, widespread implementation is still lacking, leaving large areas vulnerable to land degradation.
- **Water Conservation:** Some individuals have taken steps to harvest rainwater, reduce water wastage, and adopt efficient irrigation methods. Yet, many still lack the means or awareness to implement sustainable water management practices, exacerbating the impact of droughts and water scarcity.

2. Recommendations for Climate Action

Response Measures (Emergency Preparedness and Disaster Response)

- **Community Emergency Planning & Early Warning Systems**
 - Develop and regularly update community emergency response plans that map high-risk areas and outline evacuation procedures.
 - Establish multi-channel early warning systems, incorporating SMS alerts, radio broadcasts, and local information hubs to ensure timely climate-related alerts.
- **Evacuation and Disaster Management**
 - Design and implement clear evacuation routes and designated safe zones, ensuring accessibility for persons with disabilities (PWDs), the elderly, and those in remote areas.
 - Strengthen emergency relief coordination between government agencies, NGOs, and local communities to ensure the efficient allocation of resources, shelter, and healthcare.
 - Create and maintain an emergency response fund to provide rapid financial support for disaster relief efforts, infrastructure repairs, and emergency aid distribution.

Adaptation Strategies (Building Resilience to Climate Change)

- **Agricultural Resilience**

- Distribute drought-resistant crop varieties to farmers and provide training on climate-smart farming techniques such as crop rotation, agroforestry, soil conservation and diversification of crops.
- Promote climate-resilient agriculture by introducing sustainable farming techniques and ensuring access to efficient irrigation systems.
- Establish agro-processing to increase returns for farmers, reduce post-harvest losses, and create value-added agricultural products that provide financial stability in the face of climate variability.

- **Water Resource Management**

- Invest in large-scale water harvesting projects, including boreholes, reservoirs, and community-managed dams, to improve water availability in dry seasons.
- Encourage rainwater harvesting systems for households, institutions, and farms to enhance water security and reduce reliance on unpredictable rainfall.

- **Health System Strengthening**

- Upgrade local health facilities to manage climate-sensitive diseases such as cholera and malaria, which are worsened by extreme weather conditions.
- Deploy mobile health clinics to reach remote and disaster-affected areas, ensuring access to essential healthcare services during climate-related disruptions.

- **Transitioning to Clean Energy and Green Livelihoods**

- Promote solar-powered lighting systems and introduce clean cooking alternatives such as briquettes and energy-efficient stoves to reduce dependence on firewood and charcoal.
- Create incentives for green jobs, such as selling and distributing solar products, as a sustainable alternative to deforestation-driven income sources like charcoal burning.

Mitigation Measures (Reducing Long-Term Climate Vulnerabilities)

- **Ecosystem Protection and Sustainable Land Use Planning**

- Implement reforestation and afforestation programs, particularly restoring and protecting mangrove forests along river deltas, which act as natural flood barriers and protect biodiversity.
- Develop and enforce climate-resilient land-use plans to prevent environmental degradation, ensuring zoning policies protect agricultural land, water catchment areas, and forests from encroachment.

- **Building Climate-Resilient Infrastructure and Awareness**

- Invest in climate-resilient infrastructure, including flood-resistant roads, reinforced housing, and improved drainage systems, to minimize damage from extreme weather events.
- Strengthen community-wide climate education and awareness campaigns, equipping local populations with knowledge and tools for conservation efforts, disaster preparedness, and sustainable resource management.

Conclusion

The findings of this Climate Action Project in Yumbe District underscore the urgent need for integrated, community-driven approaches to address the region's growing climate vulnerabilities. Erratic rainfall, prolonged droughts, rising temperatures, and deforestation have significantly disrupted agriculture, water availability, and livelihoods, exacerbating food insecurity and economic instability. The presence of the Bidibidi Refugee Settlement has further intensified competition for natural resources, increasing pressure on land, forests, and water sources.

Through participatory data collection and validation sessions, this project has empowered local communities, government authorities, and humanitarian organizations to collaboratively assess climate risks and develop targeted response, adaptation, and mitigation strategies. Key recommendations include strengthening early warning systems, promoting sustainable water management, introducing climate-resilient agriculture, expanding access to renewable energy, and enforcing environmental conservation policies.

However, achieving long-term climate resilience in Yumbe will require sustained commitment from stakeholders, enhanced resource allocation, and stronger policy enforcement. Community engagement and leadership will remain central to ensuring that interventions are inclusive, context-specific, and aligned with local priorities. By fostering collaborative climate action, Yumbe District can build stronger adaptive capacities, safeguard livelihoods, and reduce vulnerabilities to future climate shocks.

This report provides a foundation for continued engagement, advocacy, and investment in climate resilience efforts. It offers a roadmap for policymakers, humanitarian organizations, and development partners to support sustainable, community-led solutions in Yumbe District.

The background is a solid blue color. Overlaid on this is a complex, abstract geometric design. A large sphere is the central focus, constructed from a network of thin, light-blue lines that form a triangular mesh. This sphere is surrounded by several concentric dashed circles. Scattered throughout the composition are numerous small dots, some of which are enclosed in small circles. The overall aesthetic is technical and digital, suggesting themes of networking, data, or global connectivity.

NETHOPE

nethope.org